

Annex 2.

EES Code for Modeling a Proposed MSSG with Equation Derivation

This annex contains the following elements:

1. The original EES code, preserving all comments exactly as they were written.
2. The full derivation of the key equations used in the heat exchangers (superheater, reheater, boiler, and preheater), preceded by a clear definition of each variable.
3. Finally, the numerical results obtained from the calculations are presented, organized into tables or an executive summary.

"!EVALUATION OF THE PROPOSED MSSG (SUPERHEATER2–SUPERHEATER1–REHEATER–EVAPORATOR–PREHEATER)"

"!Define the bookkeeping method"

turb_in = 1 "turbine inlet"

ph_in = 36 "Preheater HX inlet"

boil_in = 49 "Boiler inlet"

sh1_in = 50 "Superheater HX inlet"

sh2_in = 51 "Secondary superheater inlet"

RHC = 4 "Main steam inlet to the valve box"

RHH = 5 "Superheater inlet"

"!Original heat transfer fluid data"

T_htf_hot_sh2 = 838.7 [K]

T_htf_cold = 561.2 [K]

T_htf_hot_rh = 838.7 [K]

"!Original conditions considering a pressure drop in the boiler"

T[turb_in] = 811.15 [K]

P[turb_in] = 16700000 [Pa]

h[turb_in] = enthalpy(Steam_IAPWS, T=T[turb_in], P=P[turb_in])

P[sh1_in] = 18214192 [Pa]

T[sh1_in] = T_boil

h[sh1_in] = enthalpy(Steam_IAPWS, T=T[sh1_in], P=P[sh1_in])

P[boil_in] = 18214192 [Pa]

T_boil = t_sat(Steam_IAPWS, P=P[boil_in])

T[boil_in] = T_boil - 0.01 [K]

h[boil_in] = enthalpy(Steam_IAPWS, T=T[boil_in], P=P[boil_in])

T[ph_in] = 524.6

P[ph_in] = 18481616 [Pa]

h[ph_in] = enthalpy(Steam_IAPWS, T=T[ph_in], P=P[ph_in])

m_dot_steam = 84.84 [kg/s]

T[sh2_in] = 642 [K]

P[sh2_in] = 17554054

h[sh2_in] = enthalpy(Steam_IAPWS, T=T[sh2_in], P=P[sh2_in])

"!Reheater inlet"

P[RHC] = 4198761 [Pa]

T[RHC] = 615.4 [K]

h[RHC] = enthalpy(Steam_IAPWS, T=T[RHC], P=P[RHC])

"!Reheater outlet"

P[RHH] = 3779027 [Pa]

T[RHH] = 811.15 [K]

$h[RHH] = \text{enthalpy}(\text{Steam_IAPWS}, T=T[RHH], P=P[RHH])$

$m_dot_RH = 76.17 \text{ [kg/s]}$

"!Heat transfer fluid"

HTF\$ = 'Salt(60NaNO3_40KNO3)'

"!DESIGN UA CALCULATION FOR HEAT EXCHANGERS (REF)"

"!HTF mass flow in the secondary superheater"

$T_htf_med_sh2 = (T_htf_hot_sh2 + T_htf_cold) / 2$

$cp_htf_avg1 = \text{specheat}(\text{HTF}\$, T=T_htf_med_sh2)$ "Average specific heat of HTF between SH inlet and outlet"

$m_dot_htf1 = ((h[turb_in] - h[sh2_in]) * m_dot_steam) + ((h[sh1_in] - h[ph_in]) * m_dot_steam) / (cp_htf_avg1 * (T_htf_hot_sh2 - T_htf_cold))$ "HTF1 mass flow"

"!Heat absorbed by steam in the secondary superheater"

$q_sh2 = h[turb_in] - h[sh2_in]$ "Specific heat absorbed by water in SH"

$Q_dot_sh2 = q_sh2 * m_dot_steam$ "Heat absorbed by water in SH"

$cp_s_sh2 = \text{specheat}(\text{Steam_IAPWS}, T=((T[turb_in] + T[sh2_in]) / 2), P=P[sh2_in])$ "Specific heat of water in SH"

$C_dot_s_sh2 = cp_s_sh2 * m_dot_steam$ "Heat capacity rate of water in SH"

$T_htf_avg_sh2 = (T_htf_hot_sh2 + T_htf_boil_in) / 2$

$cp_htf_sh2 = \text{specheat}(\text{HTF}\$, T=T_htf_avg_sh2)$ "Average specific heat of HTF in SH"

$C_dot_hft_sh2 = cp_htf_sh2 * m_dot_htf1$ "Heat capacity rate of HTF in SH"

$Q_dot_htf_sh2 = C_dot_hft_sh2 * (T_htf_hot_sh2 - T_htf_boil_in)$

"!Superheater size"

$C_dot_min_sh2 = \min(C_dot_s_sh2, C_dot_hft_sh2)$ "Minimum heat capacity rate"

$C_dot_max_sh2 = \max(C_dot_s_sh2, C_dot_hft_sh2)$ "Maximum heat capacity rate"

$C_R_sh2 = C_dot_min_sh2 / C_dot_max_sh2$ "Heat capacity ratio"

$Q_dot_max_sh2 = C_dot_min_sh2 * (T_htf_hot_sh2 - T[sh1_in])$

"!Effectiveness–NTU relationship for counterflow implemented in EES HX library"

$\epsilon_{sh2} = Q_dot_sh2 / (C_dot_min_sh2 * (T_htf_hot_sh2 - T[sh2_in]))$ "Effectiveness"

"!Actual outlet temperature of the secondary superheater"

$T_htf_boil_in = T_htf_hot_sh2 - (Q_dot_sh2 / C_dot_hft_sh2)$ "Actual HTF inlet temperature to the boiler, check assumption"

"!HTF mass flow in the Reheater"

$T_htf_med_rh = (T_htf_hot_rh + T_htf_cold) / 2$

$cp_htf_avg2 = \text{specheat}(\text{HTF}\$, T=T_htf_med_rh)$

"Average specific heat of HTF between RHC inlet and RHH outlet"

$m_dot_htf2 = ((h[RHH] - h[RHC]) * m_dot_RH) + ((h[sh2_in] - h[sh1_in]) * m_dot_steam) + ((h[boil_in] - h[ph_in]) * m_dot_steam) / ((cp_htf_avg2) * (T_htf_hot_rh - T_htf_cold))$ "HTF mass flow"

"!Heat absorbed by the steam in the Reheater"

$q_RH = h[RHH] - h[RHC]$ "Specific heat absorbed by water in the RH"

$Q_dot_RH = q_RH * m_dot_RH$ "Heat absorbed by water in the RH"

$cp_s_RH = \text{specheat}(\text{Steam_IAPWS}, T=((T[RHH] + T[RHC]) / 2), P=P[RHC])$ "Specific heat of water in the RH"

$C_dot_s_RH = cp_s_RH * m_dot_RH$ "Heat capacity rate of water in the RH"

$T_htf_avg_rh = (T_htf_hot_rh + T_htf_sh1_in) / 2$

$cp_htf_RH = \text{specheat}(\text{HTF}\$, T=T_htf_avg_rh)$

"Average specific heat of HTF in the RH"

$C_dot_hft_RH = cp_htf_RH * m_dot_htf2$ "Heat capacity rate of HTF in the RH"

$Q_dot_htf_RH = C_dot_hft_RH * (T_htf_hot_rh - T_htf_sh1_in)$

"!Reheater size"

$C_dot_min_RH = \min(C_dot_s_RH, C_dot_hft_RH)$ "Minimum heat capacity rate"

$C_{dot_max_RH} = \max(C_{dot_s_RH}, C_{dot_hft_RH})$ "Maximum heat capacity rate"
 $C_{R_RH} = C_{dot_min_RH} / C_{dot_max_RH}$ "Heat capacity ratio"

$Q_{dot_max_RH} = C_{dot_min_RH} * (T_{htf_hot_rh} - T_{RHC})$

"!Effectiveness-NTU relationship for counterflow implemented using EES HX library"
 $\epsilon_{RH} = Q_{dot_RH} / (C_{dot_min_RH} * (T_{htf_hot_rh} - T_{RHC}))$ "Effectiveness"

"Alternative form"

$ARH = \epsilon_{RH} - 1$
 $BRH = \epsilon_{RH} * C_{R_RH} - 1$
 $CRH = \ln(ARH / BRH)$
 $DRH = C_{R_RH} - 1$
 $NTU_{RH} = CRH / DRH$

$UA_{RH} = NTU_{RH} * C_{dot_min_RH}$ "Heat transfer coefficient times area for RH"

"!Actual HTF outlet temperature from the Reheater"

$T_{htf_sh1_in} = T_{htf_hot_rh} - (Q_{dot_RH} / C_{dot_hft_RH})$ "Actual HTF outlet temperature to RH"

"!Heat absorbed by steam in the primary superheater"

$q_{sh1} = h_{sh2_in} - h_{sh1_in}$ "Specific heat absorbed by water in SH1"
 $Q_{dot_sh1} = q_{sh1} * m_{dot_steam}$ "Heat absorbed by water in SH1"

$cp_{s_sh1} = \text{specheat}(\text{Steam_IAPWS}, T = ((T_{sh2_in} + T_{sh1_in}) / 2), P = P_{sh1_in})$ "Specific heat of water in SH1"
 $C_{dot_s_sh1} = cp_{s_sh1} * m_{dot_steam}$ "Heat capacity rate of water in SH1"

$cp_{htf_sh1} = \text{specheat}(\text{HTF}, T = (T_{htf_sh1_in} + T_{htf_dren}) / 2)$ "Average specific heat of HTF in SH1"
 $C_{dot_hft_sh1} = m_{dot_htf2} * cp_{htf_sh1}$ "Heat capacity rate of HTF in SH1"

"!Primary superheater size"

$C_{dot_min_sh1} = \min(C_{dot_s_sh1}, C_{dot_hft_sh1})$ "Minimum heat capacity rate in SH1"
 $C_{dot_max_sh1} = \max(C_{dot_s_sh1}, C_{dot_hft_sh1})$ "Maximum heat capacity rate in SH1"
 $C_{R_sh1} = C_{dot_min_sh1} / C_{dot_max_sh1}$ "Heat capacity ratio in SH1"

"!Effectiveness-NTU relationship for counterflow implemented using EES HX library"

$Q_{dot_max_sh1} = C_{dot_min_sh1} * (T_{htf_sh1_in} - T_{sh1_in})$
 $\epsilon_{sh1} = Q_{dot_sh1} / (C_{dot_min_sh1} * (T_{htf_sh1_in} - T_{sh1_in}))$ "Effectiveness"

"Alternative form"

$Ash1 = \epsilon_{sh1} - 1$
 $Bsh1 = \epsilon_{sh1} * C_{R_sh1} - 1$
 $Csh1 = \ln(Ash1 / Bsh1)$
 $Dsh1 = C_{R_sh1} - 1$
 $NTU_{sh1} = Csh1 / Dsh1$

$UA_{sh1} = NTU_{sh1} * C_{dot_min_sh1}$ "Heat transfer coefficient times area for SH1"

"!Actual HTF outlet temperature from the primary superheater"

$T_{htf_dren} = T_{htf_sh1_in} - (Q_{dot_sh1} / C_{dot_hft_sh1})$ "Actual HTF outlet temperature from SH1"

"!Heat absorbed by water in the boiler"

$q_{boiler} = h_{sh1_in} - h_{boil_in}$ "Specific heat absorbed by water in the boiler"
 $Q_{dot_boiler} = q_{boiler} * m_{dot_steam}$ "Heat absorbed by water in the boiler"

"!Boiler size"

$cp_{htf_boiler} = \text{specheat}(\text{HTF}, T = (T_{htf_boil_in} + T_{htf_boiler_out}) / 2)$ "Average specific heat of HTF in the boiler"
 $C_{dot_hft_boiler} = m_{dot_htf1} * cp_{htf_boiler}$ "Heat capacity rate of HTF in the boiler"

$Q_{dot_htf_boiler} = C_{dot_hft_boiler} * (T_{htf_boil_in} - T_{htf_boiler_out})$

$Q_dot_max_boiler = C_dot_hft_boiler * (T_htf_boil_in - T[boil_in])$

$\epsilon_{boiler} = Q_dot_boiler / (C_dot_hft_boiler * (T_htf_boil_in - T[boil_in]))$ "Effectiveness"

$NTU_{boiler} = -\ln(1 - \epsilon_{boiler})$ "NTU for the boiler"

$UA_{boiler} = NTU_{boiler} * C_dot_hft_boiler$ "Heat transfer coefficient times area for the boiler"

"!Actual inlet temperature to the preheater"

$T_htf_boiler_out = T_htf_boil_in - (Q_dot_boiler / C_dot_hft_boiler)$ "HTF inlet temperature to the actual preheater, comment out the assumed one"

$Cp_htf_ph_in = cp('Salt(60NaNO3_40KNO3)', T = T_htf_boiler_out)$

$Cp_htf_dren = cp('Salt(60NaNO3_40KNO3)', T = T_htf_dren)$

$T_htf_ph_in = ((m_dot_htf1 * Cp_htf_ph_in * (T_htf_boiler_out)) + (m_dot_htf2 * Cp_htf_dren * (T_htf_dren))) / ((m_dot_htf1 + m_dot_htf2) * ((Cp_htf_ph_in + Cp_htf_dren) / 2))$

"!HTF mass flow in the boiler and preheater"

$m_dot_htf = m_dot_htf1 + m_dot_htf2$

"!Heat absorbed by the water in the preheater"

$q_ph = h[boil_in] - h[ph_in]$ "Specific heat absorbed by water in the preheater"

$Q_dot_ph = q_ph * m_dot_steam$ "Heat absorbed by water in the preheater"

$cp_w_ph = specheat(Steam_IAPWS, T = ((T[boil_in] + T[ph_in]) / 2), P = P[boil_in])$ "Specific heat of water in the preheater"

$C_dot_w_ph = cp_w_ph * m_dot_steam$ "Heat capacity rate of water in the preheater"

$T_htf_avg_ph = (T_htf_ph_in + T_htf_cold) / 2$

$cp_htf_ph = specheat(HTF, T = (T_htf_avg_ph))$ "Average specific heat of HTF in the preheater"

$C_dot_hft_ph = m_dot_htf * cp_htf_ph$ "Heat capacity rate of HTF in the preheater"

$Q_dot_htf_ph = C_dot_hft_ph * (T_htf_ph_in - T_htf_cold)$

"!Preheater size"

$C_dot_min_ph = \min(C_dot_w_ph, C_dot_hft_ph)$ "Minimum heat capacity rate"

$C_dot_max_ph = \max(C_dot_w_ph, C_dot_hft_ph)$ "Maximum heat capacity rate"

$C_R_ph = C_dot_min_ph / C_dot_max_ph$ "Heat capacity ratio"

$Q_dot_max_ph = C_dot_min_ph * (T_htf_ph_in - T[ph_in])$

"!Effectiveness–NTU relationship for counterflow implemented using EES HX library"

$\epsilon_{ph} = Q_dot_ph / (C_dot_min_ph * (T_htf_ph_in - T[ph_in]))$ "Effectiveness"

"Alternative form"

$A_{ph} = \epsilon_{ph} - 1$

$B_{ph} = \epsilon_{ph} * C_R_ph - 1$

$C_{ph} = \ln(A_{ph} / B_{ph})$

$D_{ph} = C_R_ph - 1$

$NTU_{ph} = C_{ph} / D_{ph}$

$UA_{ph} = NTU_{ph} * C_dot_min_ph$ "Heat transfer coefficient times area for the preheater"

"!NUMERICAL RESULTS OBTAINED"

$A_{ph} = -0.1772$

$ARH = -0.1042$

$Ash1 = -0.6272$

$boil_in = 49$

$B_{ph} = -0.6255$

$BRH = -0.5934$

$Bsh1 = -0.8737$

$C_{ph} = -1.261$

$cp_htf_avg1 = 1.521$ [kJ/kg·°K]

$cp_htf_avg2 = 1.521$ [kJ/kg·°K]

cp_htf_boiler=1.521 [kJ/kg·°K]
 Cp_htf_dren=1.522
 cp_htf_ph=1.506 [kJ/kg·°K]
 Cp_htf_ph_in=1.512
 cp_htf_RH=1.537 [kJ/kg·°K]
 cp_htf_sh1=1.526
 cp_htf_sh2=1.537
 cp_s_RH=2.330 [kJ/kg·°K]
 cp_s_sh1=13.5
 cp_s_sh2=3.558
 cp_w_ph=5.477 [kJ/kg·°K]
 CRH=-1.74
 Csh1=-0.3314
 C_dot_hft_boiler=643.7
 C_dot_hft_ph=1021
 C_dot_hft_RH=391
 C_dot_hft_sh1=388.1
 C_dot_hft_sh2=650.7
 C_dot_max_ph=1021
 C_dot_max_RH=391
 C_dot_max_sh1=1145
 C_dot_max_sh2=650.7
 C_dot_min_ph=464.6
 C_dot_min_RH=177.5
 C_dot_min_sh1=388.1
 C_dot_min_sh2=301.8
 C_dot_s_RH=177.5
 C_dot_s_sh1=1145
 C_dot_s_sh2=301.8
 C_dot_w_ph=464.6
 C_R_ph=0.4551
 C_R_RH=0.4539
 C_R_sh1=0.3389
 C_R_sh2=0.4639
 Dph=-0.5449
 DRH=-0.5461
 Dsh1=-0.6611
 epsilon_boiler=0.8592 [-]
 epsilon_ph=0.8228 [-]
 epsilon_RH=0.8958 [-]
 epsilon_sh1=0.3728
 epsilon_sh2=0.9981
 HTF\$='Salt(60NaNO3_40KNO3)'
 m_dot_htf=677.7 [kJ/s]
 m_dot_htf1=423.3 [kg/s]
 m_dot_htf2=254.4 [kg/s]
 m_dot_RH=76.17 [kg/s]
 m_dot_steam=84.84 [kg/s]
 NTU_boiler=1.961
 NTU_ph=2.315
 NTU_RH=3.186
 NTU_sh1=0.5013
 ph_in=36
 q_boiler=759.6
 Q_dot_boiler=64448 [kJ/s]
 Q_dot_htf_boiler=64448 [kJ/s]